

ABSTRACT

In today's globalized era, integrating smart technologies into disaster management has become essential for effective public safety and operational efficiency. This research seeks to bridge emergency communication and response effectiveness, supporting Candon City in its initiative to become a SMART City, making it a safer, more resilient, and technology-facilitated urban center. The study utilized descriptive and developmental methods, incorporating surveys and software development. The researchers employed the Agile programming methodology during the software development process and used random sampling to select 180 respondents for evaluating software usability.

Candon City's current emergency response mechanisms remain largely traditional and fragmented, resulting in delays, limited accessibility, and reduced accountability, highlighting the need for a centralized system to enable more effective and efficient emergency response management. Assessed using ISO 9126-1 standards, the developed system achieved an overall mean score of 4.81, reflecting very high usability and outstanding performance across all dimensions of software quality.

The adoption of the developed system is recommended to enhance real-time reporting, coordination, and accountability in emergency management in Candon City. Continuous assessment and iterative improvements should be conducted to ensure the developed system remains responsive to evolving user needs and technological advancements.

Keywords: *agile programming, centralized system, emergency management, public safety, smart city*